

SK Digital Solutions WEEK II

Part I

Nature of Business:

SK Digital Solutions is a newly established software development company that provides technology solutions and IT services to businesses. The company focuses on developing software applications, providing technical support, and delivering reliable technology solutions to meet the needs of its clients.

Company Vision:

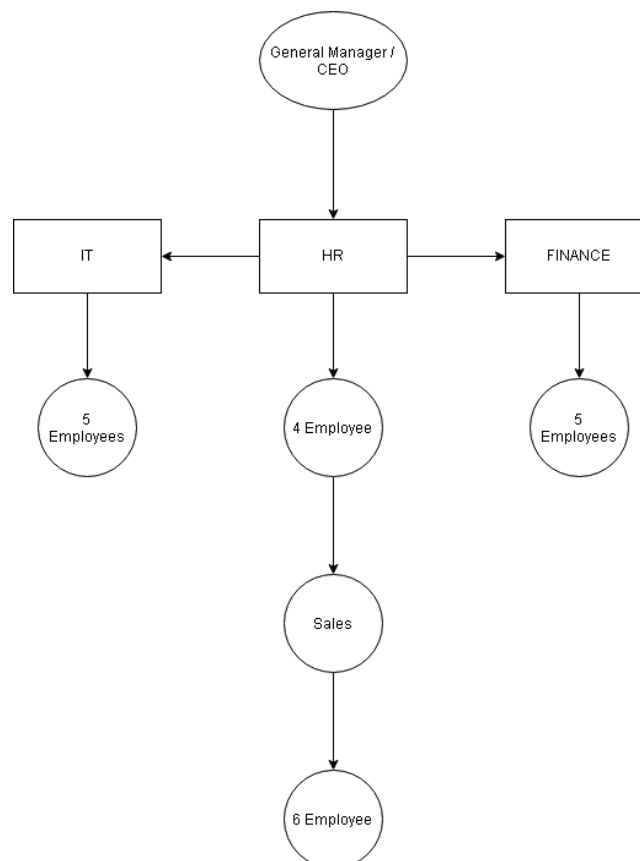
To become a trusted and innovative technology company that delivers reliable, secure, and high-quality software solutions to businesses and organizations.

Company Mission:

SK Digital Solutions is committed to providing effective software and IT solutions through the use of modern technologies, skilled employees, secure infrastructure, and excellent customer service. The company aims to help its clients improve their business operations through practical and reliable technology

Office Location:

SK Digital Solution
017 Sitio Uno, Brgy. Lambac
Pagsanjan, Laguna



Department Functions:

IT department - 5 employees = 25%

HR Department - 4 employees = 20%

Finance Department - 5 employees = 25%

Sales Department - 6 employees = 30%

Total 20 = 100%

Part II - Enterprise Hardware Inventory

Hardware Inventory Table

Asset ID	Hardware	Quantity	Department	Purpose
H01	Desktop Computer	16	IT, HR, Finance, Sales	Primary workstation for employees performing daily office and development tasks
H02	Laptop Computer	4	IT, HR, Finance, Sales	Provides mobility for employees who need to work in different areas or attend meetings
H03	Server	1	IT	Provides centralized services, file storage, applications, and system administration
H04	Router	1	IT	Connects the company's internal network to the Internet and manages network traffic
H05	Managed Network Switch	1	IT	Connects wired computers, server, printers, NAS, and other network devices
H06	Network Printer	2	HR, Finance, General Office	Provides shared printing for documents, reports, and administrative requirements
H07	UPS	2	IT	Protects the server and network equipment from power interruptions and fluctuations
H08	Wireless Access Point	3	All Departments	Provides wireless network connectivity throughout the office
H09	NAS Storage	1	IT	Provides centralized network storage and backup capacity
H10	External Backup Drive	2	IT	Provides additional offline backup and recovery storage
H11	Monitor	20	All Departments	Provides display devices for employee workstations

Hardware Quantity Justification

Desktop Computers – 16 Units

Sixteen desktop computers are proposed for employees whose work is primarily performed at their assigned office workstations. Desktop computers provide a practical solution for regular office work and software development.

Laptop Computers – 4 Units

Four laptops are proposed to provide mobility for selected employees. Laptops can be used for meetings, presentations, remote work, troubleshooting, and other situations where a portable computer is useful.

Together, the **16 desktops and 4 laptops provide 20 primary computers**, corresponding to the company's 20 employees.

Server – 1 Unit

One dedicated server is proposed for centralized IT services. It will be managed by the IT department and can provide services such as centralized file storage, internal applications, user-related services, and other server functions required by the organization.

Router – 1 Unit

One router is required to connect the company's internal network to the Internet and manage traffic between the internal network and external networks.

Managed Network Switch – 1 Unit

One managed switch provides wired connectivity between employee computers, the server, printers, NAS storage, and other network equipment. A managed switch also provides room for better network management as the organization grows.

Network Printers – 2 Units

Two network printers are proposed so employees can share printing resources. One can primarily support HR and Finance, while the second can serve the general office and other departments.

UPS – 2 Units

Two UPS units are recommended to protect critical infrastructure from power interruptions and electrical fluctuations. The UPS units can be assigned to the server and core networking equipment.

Wireless Access Points – 3 Units

Three wireless access points are proposed to provide wireless coverage throughout the company's single office floor. Multiple access points can provide better coverage and accommodate employee wireless devices.

NAS Storage – 1 Unit

One NAS device provides centralized network storage and can also be used as a destination for backups. This gives the company an additional storage location separate from the primary server.

External Backup Drives – 2 Units

Two external backup drives provide additional backup storage. Having two drives allows the company to maintain separate backup copies and keep one backup drive offline when appropriate.

Monitors – 20 Units

Twenty monitors are proposed so that each of the company's 20 employees has a dedicated display for their primary workstation.

Software	Version	License	Purpose
Win 11 Pro	Current Supported Version	Commercial	Primary operating system for employee desktop and laptop computers
Ubuntu Server	Current LTS Version	Open Source	Operating system for the company's server
Microsoft Office	Microsoft 365 Apps / Current Version	Subscription	Word processing, spreadsheets, presentations, email, and office productivity
Visual Studio Code	Current Stable Version	Free	Source-code editing and software development
Git	Current Stable Version	Open Source	Source-code version control and collaboration
GitHub Desktop	Current Version	Free	Graphical interface for managing Git repositories
VirtualBox	Current Version	Open Source	Creates and manages virtual machines for testing and development
Google Chrome	Current Stable Version	Free	Web browsing and testing web-based applications
Microsoft Defender	Included with Windows	Included	Malware protection and endpoint security
AnyDesk	Current Business Version	Commercial	Remote technical support and troubleshooting
7-Zip	Current Stable Version	Open Source	File compression, extraction, and archive management

Software Purpose and Justification

1. Windows 11 Pro

Windows 11 Pro will serve as the primary operating system for the company's desktop and laptop computers. It provides the environment employees need to perform office work, software development, communication, and other business activities.

The Pro edition is appropriate for a business environment because it provides features intended for professional and organizational use.

2. Ubuntu Server

Ubuntu Server will be installed on the company's dedicated server. It provides a stable server operating system for hosting internal services and managing server resources.

It is particularly appropriate for a software development company because Linux is widely used in server and development environments.

3. Microsoft Office

Microsoft Office will be used by employees for common business activities such as creating documents, preparing spreadsheets, producing presentations, and managing administrative information.

The Finance and HR departments can use spreadsheet and document applications for reports and records, while Sales can use them for presentations and business documents.

4. Visual Studio Code

Visual Studio Code will primarily support the Information Technology department. Developers can use it to write, edit, and manage source code for software and web applications.

It is a lightweight and flexible development environment suitable for different programming languages and development projects.

5. Git

Git provides version control for software development projects. It allows developers to track changes to source code, create branches, and maintain different versions of a project.

This is essential for the IT department because ABC Startup Solutions operates as a software development company.

6. GitHub Desktop

GitHub Desktop provides a graphical interface for Git and GitHub workflows. It can make repository management easier for employees who prefer a graphical interface instead of using Git entirely through the command line.

It can be used to clone repositories, commit changes, manage branches, and synchronize project files.

7. VirtualBox

VirtualBox allows IT personnel and developers to create virtual machines on their computers.

The IT department can use virtual machines to test different operating systems, applications, configurations, and server environments without affecting the company's production infrastructure.

8. Google Chrome

Google Chrome will be used for Internet access, web-based business applications, research, communication, and testing web applications developed by the IT department.

Because the company develops software, having a modern web browser is particularly useful for testing websites and web applications.

9. Microsoft Defender

Microsoft Defender provides security protection for Windows computers. It helps protect employee workstations from malware and other common security threats.

Keeping endpoint security enabled and updated is an important part of protecting the company's systems and data.

10. AnyDesk

AnyDesk can be used by authorized IT personnel to provide remote technical support.

Instead of physically visiting every workstation, an IT technician can remotely troubleshoot certain software and configuration problems when remote support is appropriate and permitted by company policy.

11. 7-Zip

7-Zip provides file compression and extraction capabilities. Employees can use it to compress large files, extract downloaded archives, and manage compressed project files.

It is also useful for the IT department when handling software packages, backups, and archived files.

Part IV - Enterprise Network Inventory

Asset ID	Network Equipment	Quantity	Specification / Description	Purpose
N01	ISP Modem/ONT	1	Business-grade fiber modem/ONT	Provides the connection between the ISP and the company's network
NO2	Router	1	Business-class Gigabit router	Routes traffic between the internal network and the Internet
NO3	Firewall	1	Dedicated business firewall	Protects the internal network from unauthorized external access
NO4	Managed Switch	1	24-port Gigabit managed switch	Connects computers, server, printers, NAS, and other wired devices
NO5	Wireless Access Point	3	Business-grade dual-band access points	Provides wireless network access throughout the

				office
NO6	Patch Panel	1	24-port CAT6 patch panel	Organizes and terminates Ethernet connections in the network cabinet
NO7	CAT6 Ethernet Cable	1	Bulk CAT6 cable, approximately 305 meters	Provides wired network connections between devices and the patch panel
NO8	RJ45 Connectors	100	CAT6-compatible RJ45 connectors	Used to terminate and repair Ethernet cables

1. ISP Modem/ONT

The ISP modem or optical network terminal (ONT) provides the physical connection between the Internet service provider and ABC Startup Solutions' network. For a fiber-based business Internet connection, an ONT would normally be used to terminate the fiber connection.

The device will connect to the company's router and provide Internet connectivity for employees and business systems.

2. Router

The router manages communication between the company's internal network and external networks, including the Internet.

The router will be positioned after the ISP modem/ONT and before the internal security and switching infrastructure.

3. Firewall

The firewall provides an additional security layer for the company's network. It helps control traffic entering and leaving the internal network and can be configured to block unauthorized connections.

The firewall is particularly important because the company will store business information, employee data, software projects, and other potentially sensitive information.

4. Managed Switch

A 24-port managed switch is proposed for the initial network. It will provide wired connectivity for employee computers and infrastructure devices.

The switch can connect:

- Employee computers
- Server
- NAS
- Network printers
- Wireless access points

- Other network equipment

The 24-port capacity provides connectivity for the current organization while leaving some ports available for future expansion.

5. Wireless Access Points

Three wireless access points are proposed to provide wireless coverage throughout the company's single office floor.

They can support employee laptops and other authorized wireless devices. Multiple access points also help distribute wireless clients rather than placing the entire organization's wireless traffic on one device.

6. Patch Panel

A 24-port CAT6 patch panel will organize the Ethernet cable connections from the office's network outlets.

The patch panel provides a structured and professional method of managing network cabling and makes troubleshooting or replacing connections easier.

7. CAT6 Ethernet Cable

Approximately 305 meters of CAT6 cable is recommended for the initial installation. The exact amount should be confirmed after measuring the actual office layout.

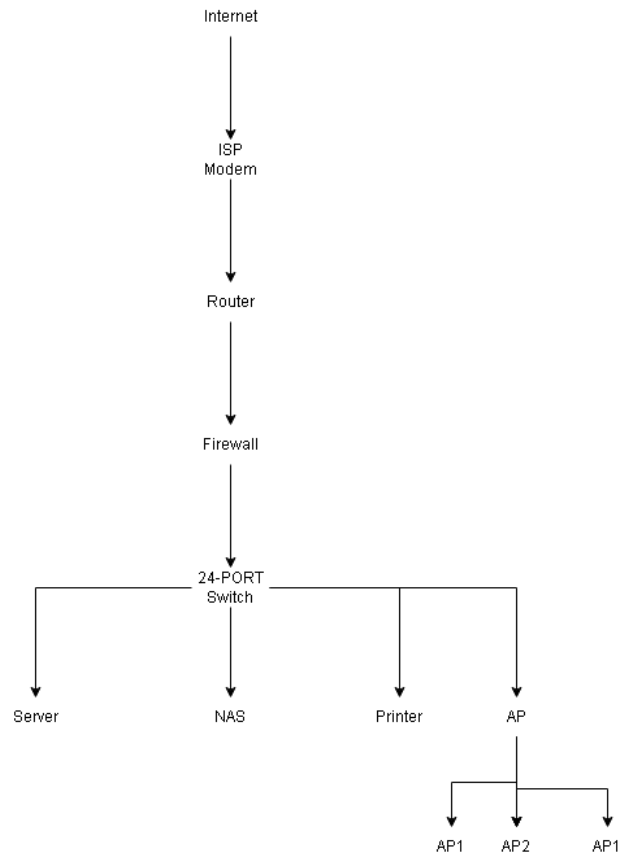
CAT6 cabling will be used for wired connections between the patch panel, switch, computers, server, printers, access points, and other network devices.

8. RJ45 Connectors

RJ45 connectors are required for terminating Ethernet cables and for future cable repairs or additional network connections.

A quantity of 100 provides spare connectors for installation, maintenance, and future expansion.

Part V - Network Layout:



Part VI - System Administration Roles

Helpdesk Technician

Responsibilities

A Helpdesk Technician is responsible for providing first-level technical support to employees. Their responsibilities include:

- Receiving and documenting user technical issues.
- Troubleshooting hardware and software problems.
- Installing and configuring computers and applications.
- Assisting users with account and password problems.
- Troubleshooting basic network connectivity issues.
- Providing remote technical assistance when appropriate.
- Escalating complex problems to specialized IT personnel.
- Maintaining records of reported and resolved issues.

Skills

A Helpdesk Technician should have:

- Hardware and software troubleshooting skills.
- Basic networking knowledge.
- Windows operating-system knowledge.
- Problem-solving skills.
- Communication and customer-service skills.
- Documentation skills.

- Basic understanding of cybersecurity.

Common Tools

Common tools include:

- Helpdesk/ticketing systems.
- Remote support software.
- Windows administrative tools.
- Hardware diagnostic utilities.
- Command Prompt and PowerShell.
- Antivirus and endpoint-security tools.

Certifications

Possible certifications include:

- CompTIA A+
 - CompTIA Network+
 - Microsoft-related certifications
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Network Administrator

Responsibilities

A Network Administrator is responsible for maintaining the organization's network infrastructure. Their responsibilities include:

- Configuring and maintaining routers and switches.
- Managing wired and wireless network connectivity.
- Monitoring network performance.
- Troubleshooting network connectivity problems.
- Managing IP addressing and network services.
- Configuring and maintaining network security controls.
- Maintaining network documentation.
- Assisting with firewall configuration and network expansion.

Skills

A Network Administrator should have:

- TCP/IP networking knowledge.
- Routing and switching skills.
- Network troubleshooting skills.
- Wireless networking knowledge.
- Network security knowledge.
- Understanding of VLANs and network segmentation.
- Knowledge of network monitoring.

Common Tools

Common tools include:

- Network-device management interfaces.
- Ping.
- Traceroute.
- Wireshark.
- SSH.
- Network monitoring software.
- Command-line networking utilities.

Certifications

Possible certifications include:

- CompTIA Network+
 - Cisco CCNA
 - CompTIA Security+
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Linux System Administrator

Responsibilities

A Linux System Administrator manages and maintains Linux-based servers. Their responsibilities include:

- Installing and configuring Linux servers.
- Managing users and permissions.
- Installing and updating server software.
- Monitoring server performance.
- Managing system services.
- Managing server storage.
- Performing system backups.
- Troubleshooting server problems.
- Maintaining Linux system security.

Skills

A Linux System Administrator should have:

- Linux command-line skills.
- File and directory management knowledge.
- User and permission management skills.
- Networking knowledge.
- Server administration skills.
- Shell scripting knowledge.
- Troubleshooting and security skills.

Common Tools

Common Linux administration tools include:

- SSH

- Bash
- apt
- systemctl
- journalctl
- top
- df
- grep

Certifications

Possible certifications include:

- CompTIA Linux+
 - Linux Foundation certifications
 - Red Hat certifications
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Cloud Administrator

Responsibilities

A Cloud Administrator manages an organization's cloud-based infrastructure and services. Their responsibilities include:

- Managing cloud computing resources.
- Configuring virtual machines.
- Managing cloud storage.
- Managing cloud user accounts and access permissions.
- Monitoring cloud resources.
- Implementing cloud backup procedures.
- Assisting with cloud security.
- Monitoring cloud resource usage and costs.
- Supporting the migration and deployment of applications to cloud environments.

Skills

A Cloud Administrator should have:

- Cloud computing knowledge.
- Networking knowledge.
- Identity and access management skills.
- Virtualization knowledge.
- Cloud security knowledge.
- Troubleshooting skills.
- Basic automation and scripting skills.

Common Tools

Examples of cloud administration tools include:

- Microsoft Azure Portal.

- AWS Management Console.
- Google Cloud Console.
- Cloud command-line tools.
- Cloud monitoring and management tools.

Certifications

Possible certifications include:

- Microsoft Azure Administrator certification.
 - AWS Certified Cloud Practitioner.
 - AWS Certified Solutions Architect.
 - Google Cloud certifications.
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How the Four Professionals Work Together

The four professionals work together to maintain the organization's complete IT environment. The **Helpdesk Technician** serves as the first point of contact for employees and handles common hardware, software, and user-support issues. When an issue involves network connectivity or network infrastructure, the **Network Administrator** investigates and resolves it. The **Linux System Administrator** manages Linux servers and ensures that server-based services remain available and secure. The **Cloud Administrator** manages cloud resources, services, storage, and access. These professionals communicate with one another, share technical documentation, escalate problems when necessary, and coordinate infrastructure changes to maintain reliable IT services for the entire organization.

Part VII - Infrastructure Recommendations

PART 7 – Infrastructure Recommendations

1. Internet Provider

SK Digital Solutions should use a **business-grade fiber Internet connection** with approximately **300–500 Mbps** speed. A backup Internet provider can be added in the future for redundancy.

2. Server Specifications

Use a dedicated server with:

- Server-grade processor
- **32 GB ECC RAM**
- **2 × 1 TB SSD with RAID 1**
- Gigabit Ethernet
- **Ubuntu Server LTS**
- UPS protection

This provides reliable centralized services and storage for the company.

3. Backup Strategy

Use the **3-2-1 backup strategy**:

- 3 copies of important data
- 2 different storage locations/media
- 1 offline or off-site copy

Use the server, NAS, and external backup drives, and regularly test data restoration.

4. Security Recommendations

SK Digital Solutions should:

- Use a firewall.
- Keep systems and software updated.
- Enable endpoint security.
- Use strong authentication and MFA.
- Apply least-privilege access.
- Protect server and network equipment physically.
- Perform regular backups.
- Provide cybersecurity awareness training.

5. Antivirus

Use **Microsoft Defender** on Windows computers and keep it updated. It should be supported by regular updates and safe computing practices.

6. Password Policy

Employees should:

- Use passwords/passphrases of at least **12 characters**.
- Avoid easily guessed information.
- Never share or reuse passwords.
- Use MFA whenever available.
- Use a password manager when appropriate.
- Use separate administrator accounts.
- Disable accounts when employees leave.

7. Expansion Plan

The infrastructure should allow for future growth. As SK Digital Solutions expands, it can add:

- More computers and network ports
- Additional access points
- Additional storage
- VLANs for department separation
- A backup Internet connection
- Additional server resources
- Cloud services and monitoring

PART 8 – Personal Reflection

This Week 2 project taught me that proper IT infrastructure planning is an important part of becoming a System Administrator. I learned how to analyze the needs of a company and identify the appropriate hardware, software, networking equipment, and security requirements. For SK Digital Solutions, I learned how to plan an infrastructure that supports 20 employees across the IT, HR, Finance, and Sales departments. I also learned the importance of documenting IT resources through hardware, software, and network inventories.

The most challenging task for me was designing the enterprise network infrastructure. I needed to understand how the Internet, ISP modem, router, firewall, switch, server, wireless access points, printers, and different departments should be connected. Creating a logical network structure helped me understand how different devices work together to provide reliable connectivity throughout an organization.

Planning is important before deployment because it helps prevent mistakes, unnecessary expenses, and technical problems. Before purchasing equipment or installing systems, a System Administrator needs to understand the organization's requirements and determine the appropriate resources. Proper planning also helps ensure that security, backup, network connectivity, and future expansion are considered from the beginning. A well-planned infrastructure can be easier to maintain, troubleshoot, and upgrade.

This project will help me become a better System Administrator by improving my technical planning, problem-solving, networking, and documentation skills. It also taught me that system administration is not only about fixing problems after they occur. A good System Administrator should be proactive and identify possible problems before they affect the organization. Through this project, I gained a better understanding of how hardware, software, networking, security, and backup strategies work together as one IT infrastructure.

Overall, the project gave me practical experience in creating an enterprise infrastructure plan for a startup company. The knowledge I gained from planning the infrastructure of SK Digital Solutions will help me become more organized, technically prepared, and capable of designing and maintaining reliable IT environments in the future.